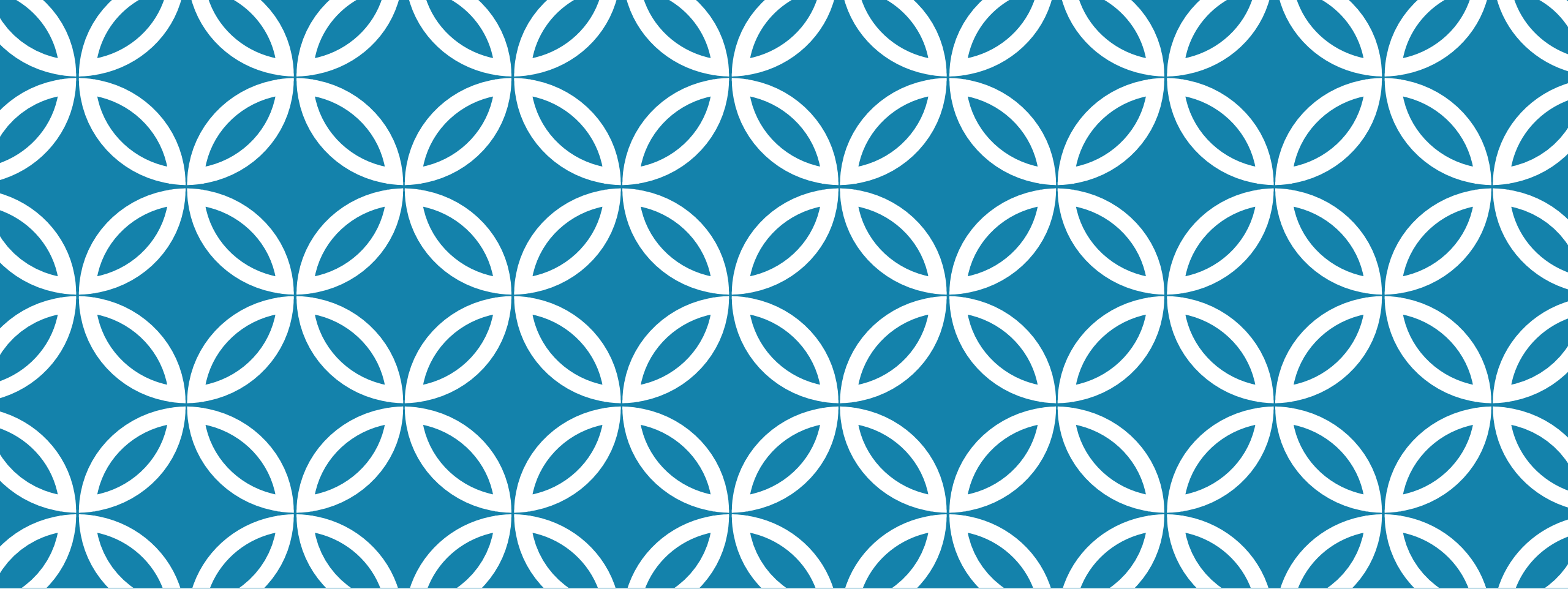




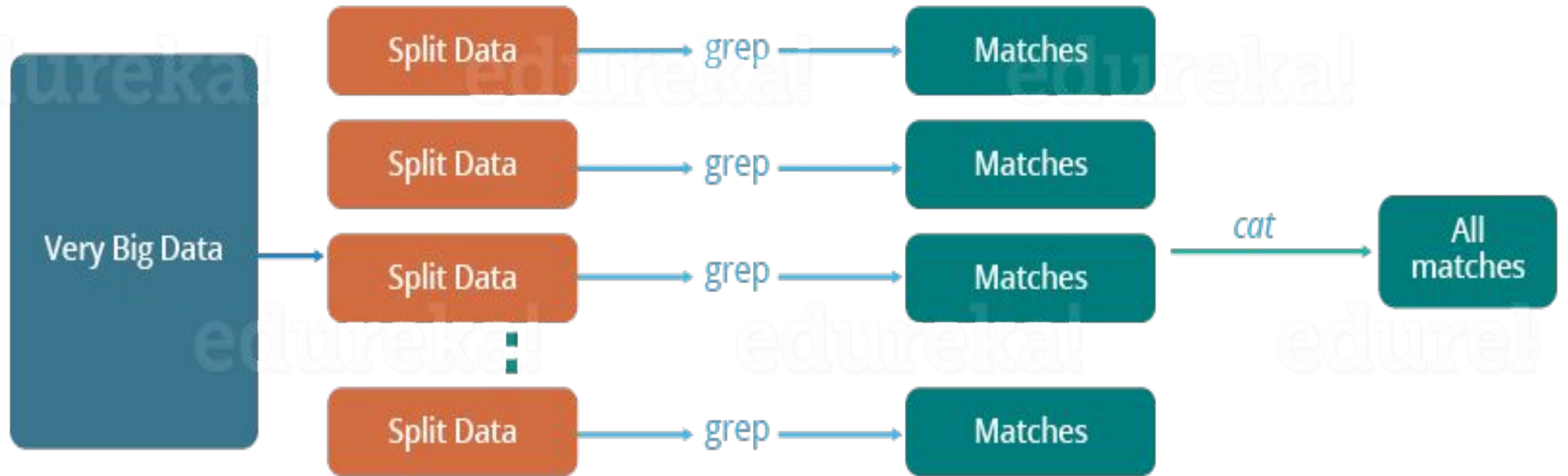
MAPREDUCE

Chapter 3

MAPREDUCE

- Traditional Way for parallel and distributed processing
- What is MapReduce?
- MapReduce Advantages

The Traditional Way



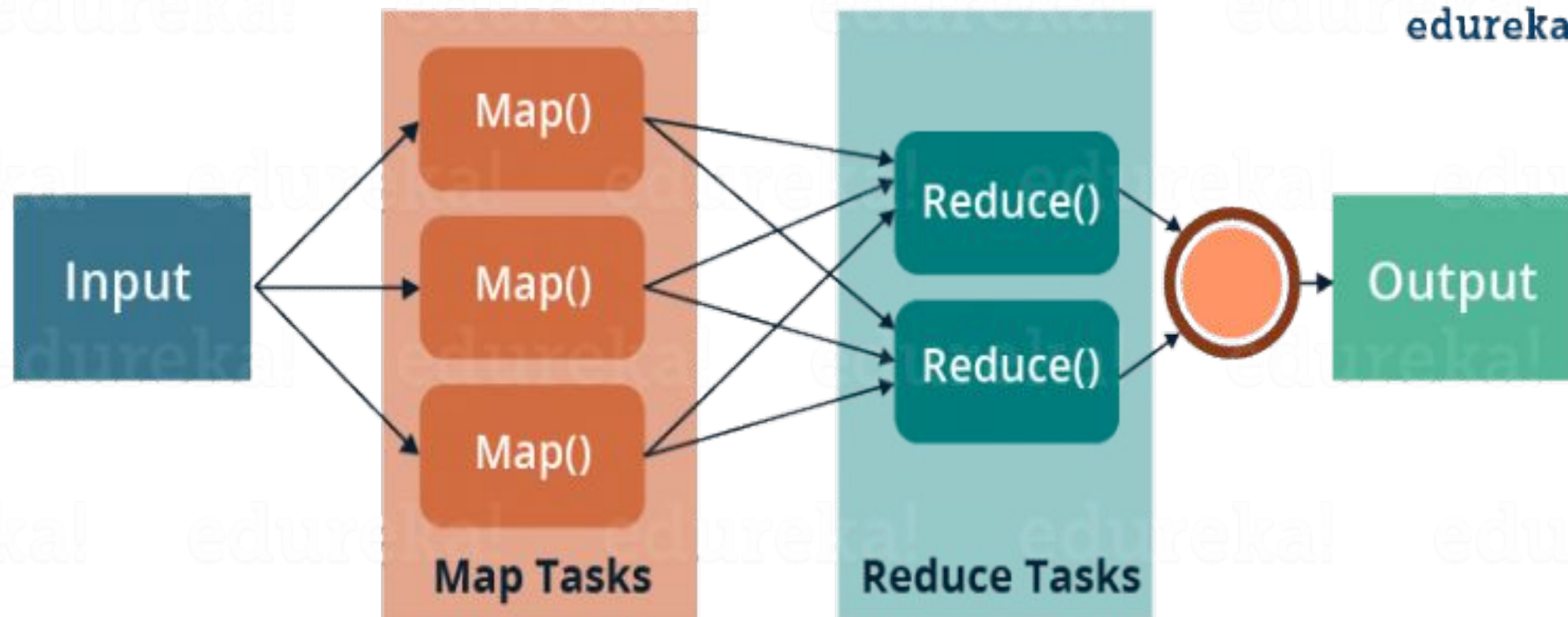
LET US LOOK AT THE CHALLENGES ASSOCIATED WITH THIS TRADITIONAL APPROACH:

- ❑ **Critical path problem:** It is the amount of time taken to finish the job without delaying the next milestone or actual completion date. So, if, any of the machines delay the job, the whole work gets delayed.
- ❑ **Reliability problem:** What if, any of the machines which are working with a part of data fails? The management of this failover becomes a challenge.
- ❑ **Equal split issue:** How will I divide the data into smaller chunks so that each machine gets even part of data to work with. In other words, how to equally divide the data such that no individual machine is overloaded or underutilized.
- ❑ **Single split may fail:** If any of the machines fail to provide the output, I will not be able to calculate the result. So, there should be a mechanism to ensure this fault tolerance capability of the system.
- ❑ **Aggregation of the result:** There should be a mechanism to aggregate the result generated by each of the machines to produce the final output.

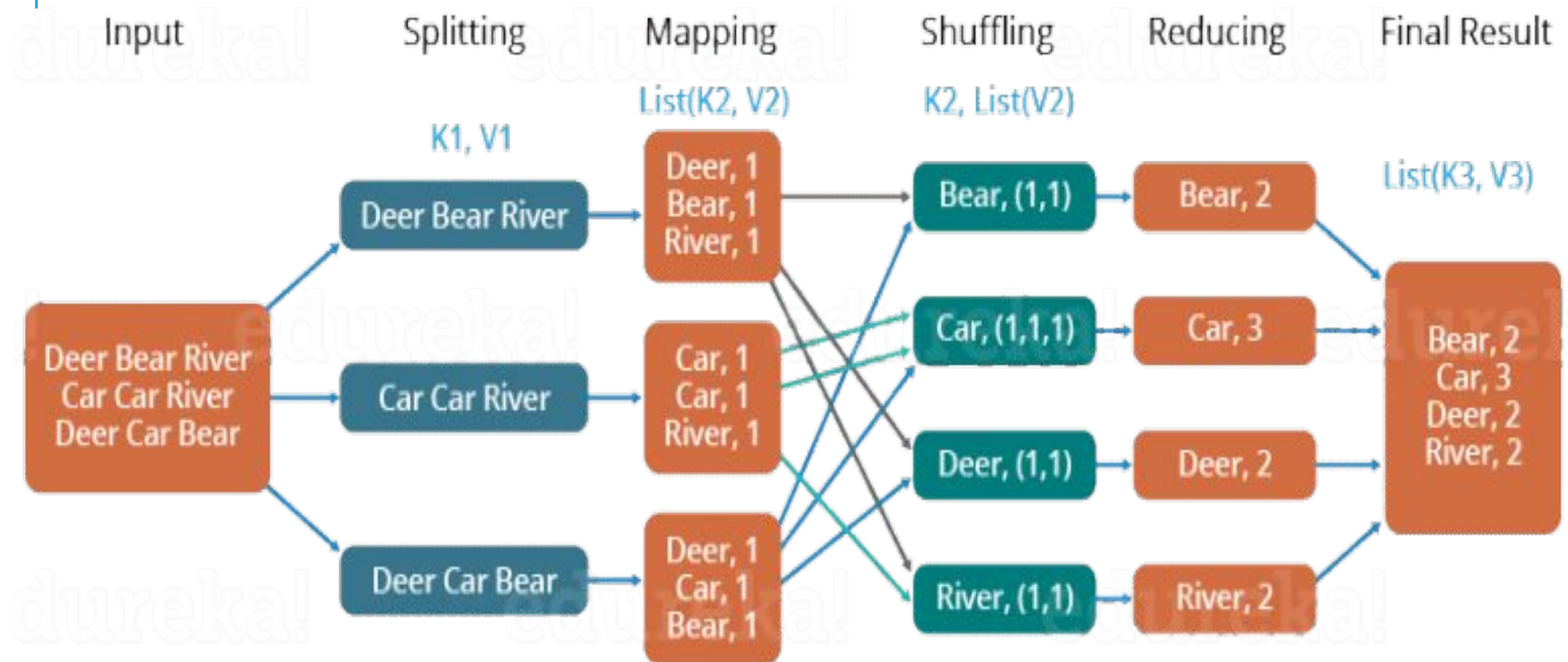
MAPREDUCE

MapReduce is a programming framework that allows us to perform distributed and parallel processing on large data sets in a distributed environment.

- ❑ MapReduce consists of two distinct tasks — Map and Reduce.
- ❑ As the name MapReduce suggests, reducer phase takes place after the mapper phase has been completed.
- ❑ So, the first is the map job, where a block of data is read and processed to produce key-value pairs as intermediate outputs.
- ❑ The output of a Mapper or map job (key-value pairs) is input to the Reducer.
- ❑ The reducer receives the key-value pair from multiple map jobs.
- ❑ Then, the reducer aggregates those intermediate data tuples (intermediate key-value pair) into a smaller set of tuples or key-value pairs which is the final output.



The Overall MapReduce Word Count Process



EXAMPLE

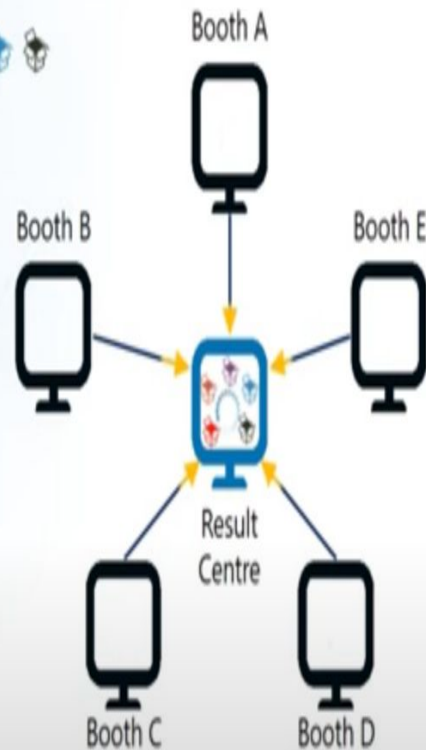
Election Votes Counting – Traditional Way

edureka!

Data →

Counting – Traditional Approach

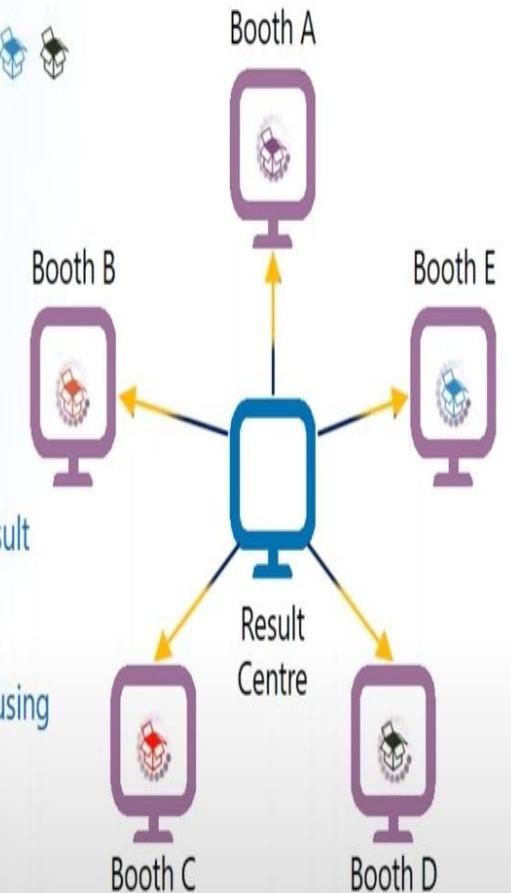
- Votes are moved to Result Centre for counting
- Moving all the votes to Centre is costly
- Result Centre is over-burdened
- Counting takes time



Counting – MapReduce Approach

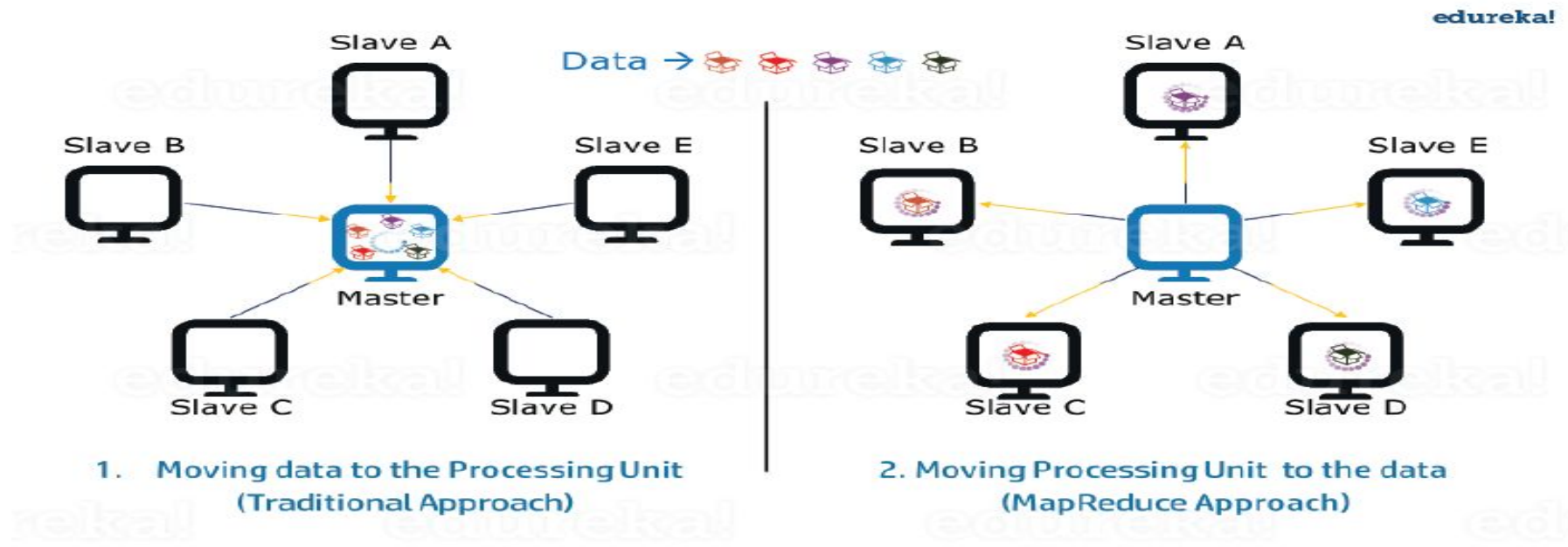
- Votes are counted at individual booths
- Booth-wise results are sent back to the result centre
- Final Result is declared easily and quickly using this way

Votes →

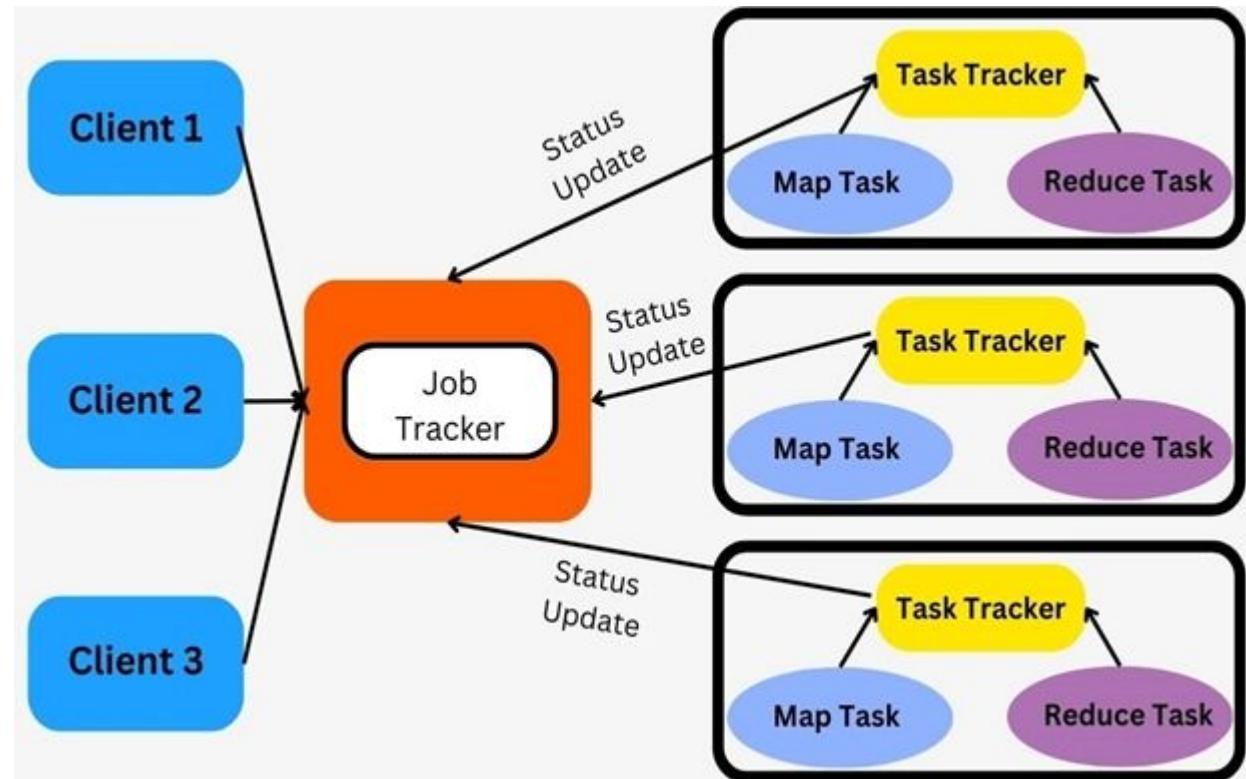


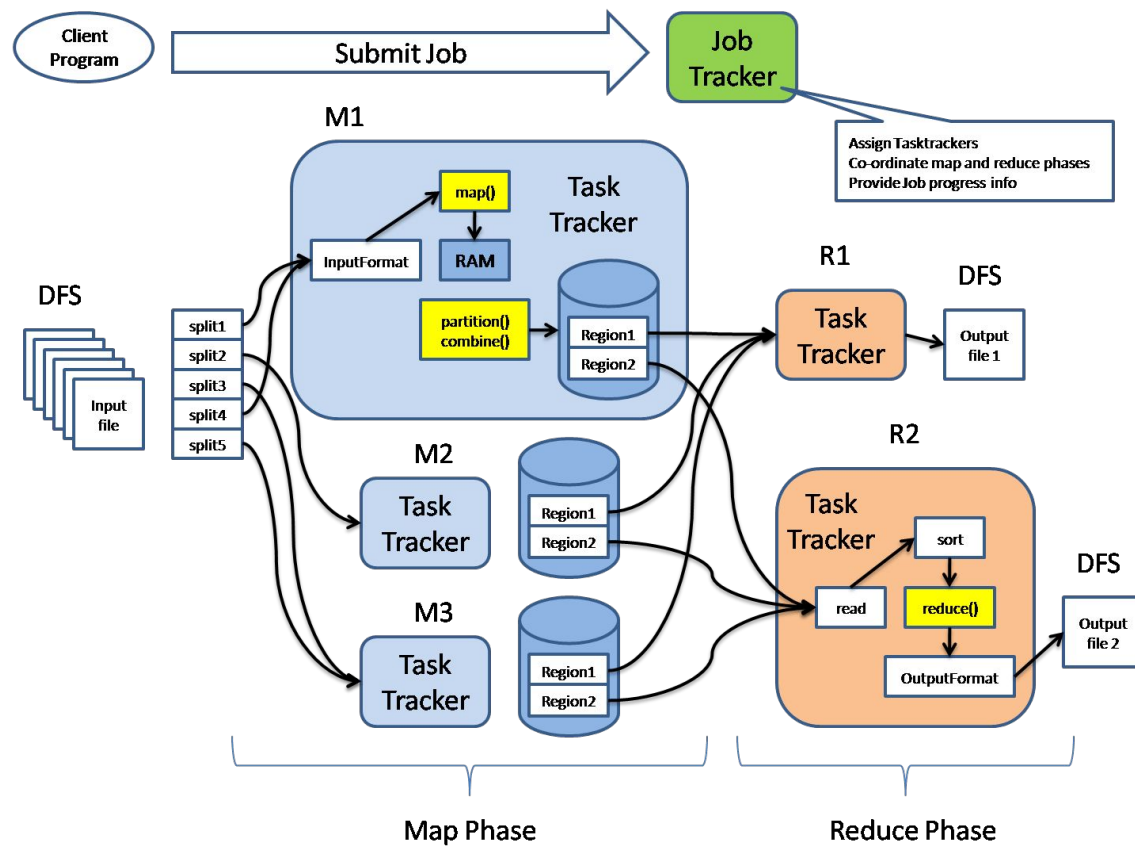
ADVANTAGES OF MAPREDUCE

- Parallel Processing
- Data Locality



MAPREDUCE ARCHITECTURE





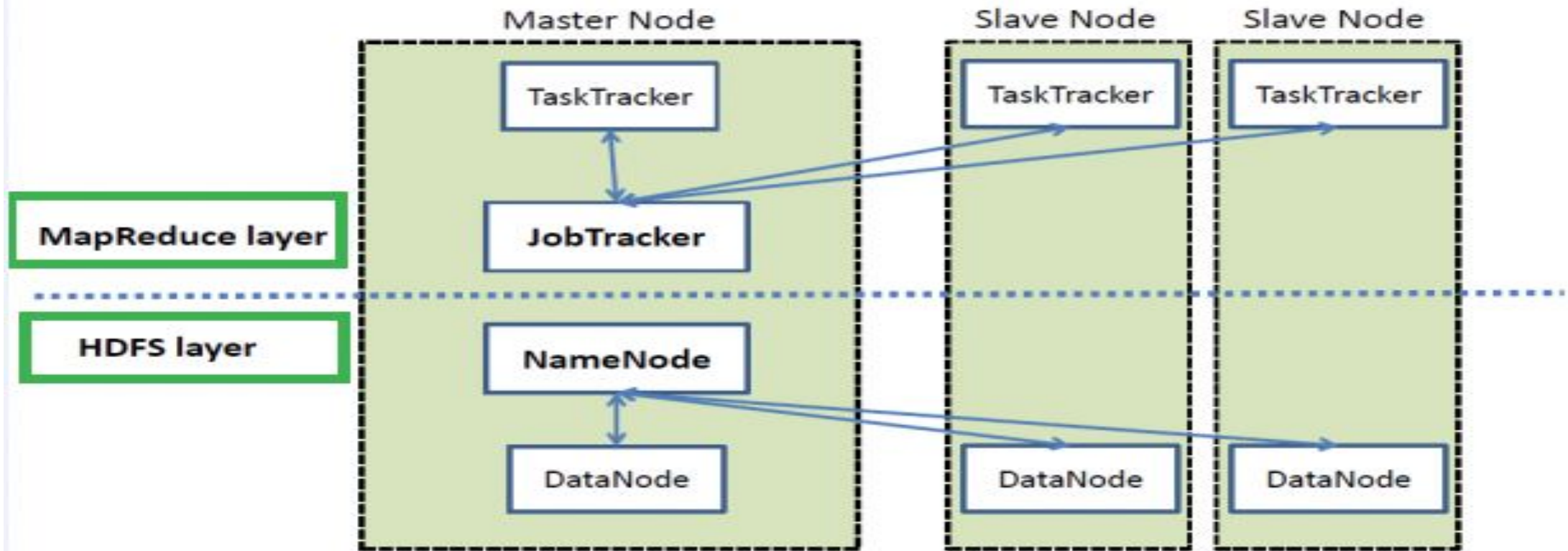
Use cases for Hadoop MapReduce applications

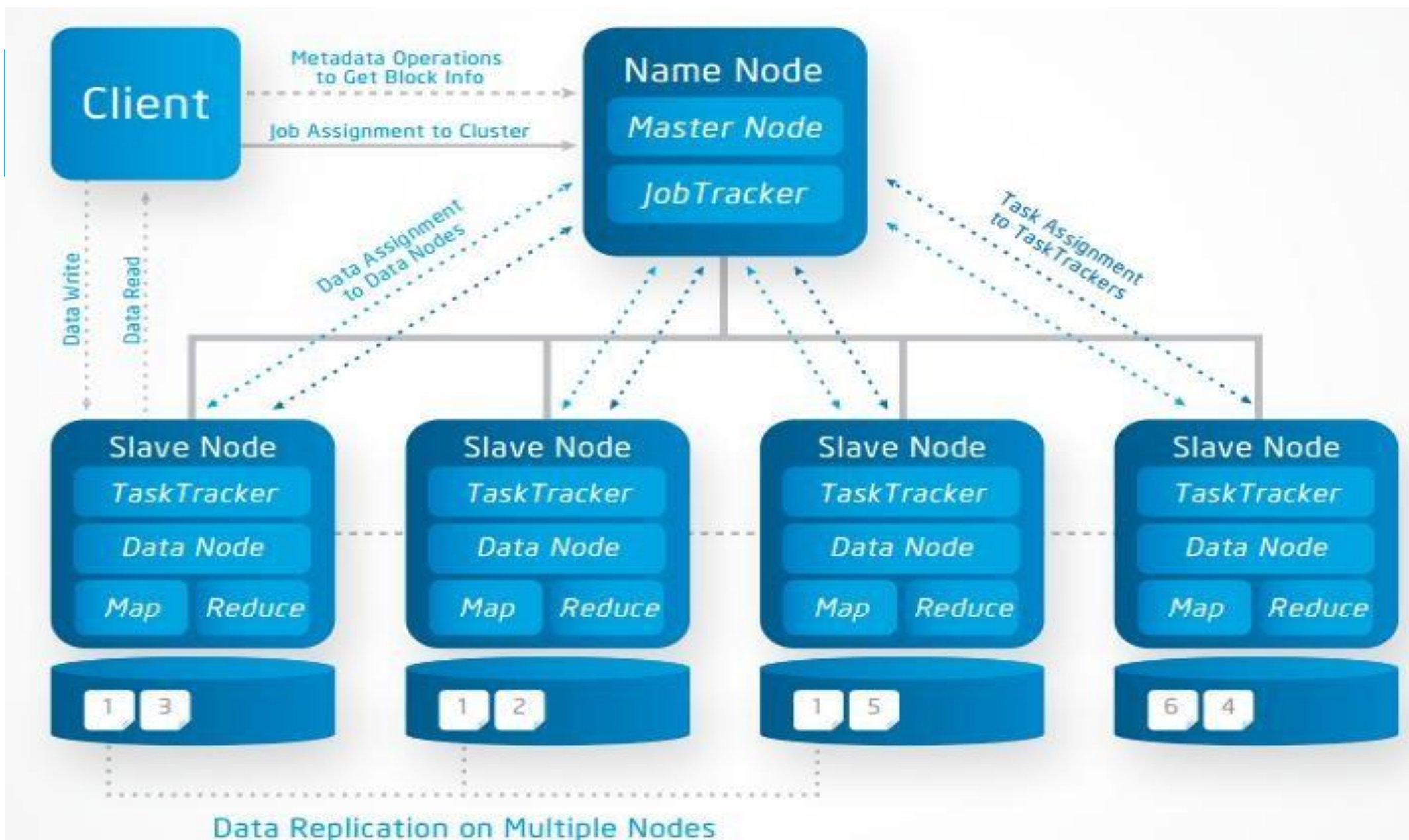


WORKING OF JOB AND TASK TRACKERS

- Each job is divided into multiple tasks, which are executed on multiple data nodes in a cluster.
- The Job tracker schedules the tasks to run on different data nodes.
- The task tracker executes the task assigned to it by the job tracker. Task trackers are present on every data node executing a part of the job.
- Task Tracker is also responsible for sending status updates to the job tracker.
- Task Tracker periodically sends heartbeat signals to the Job tracker to notify it about the system's current state.
- Hence, the Job tracker tracks the progress of each job with the help of task trackers. In case of task failure, the job tracker reschedules the task to a different task tracker.

High Level Architecture of Hadoop



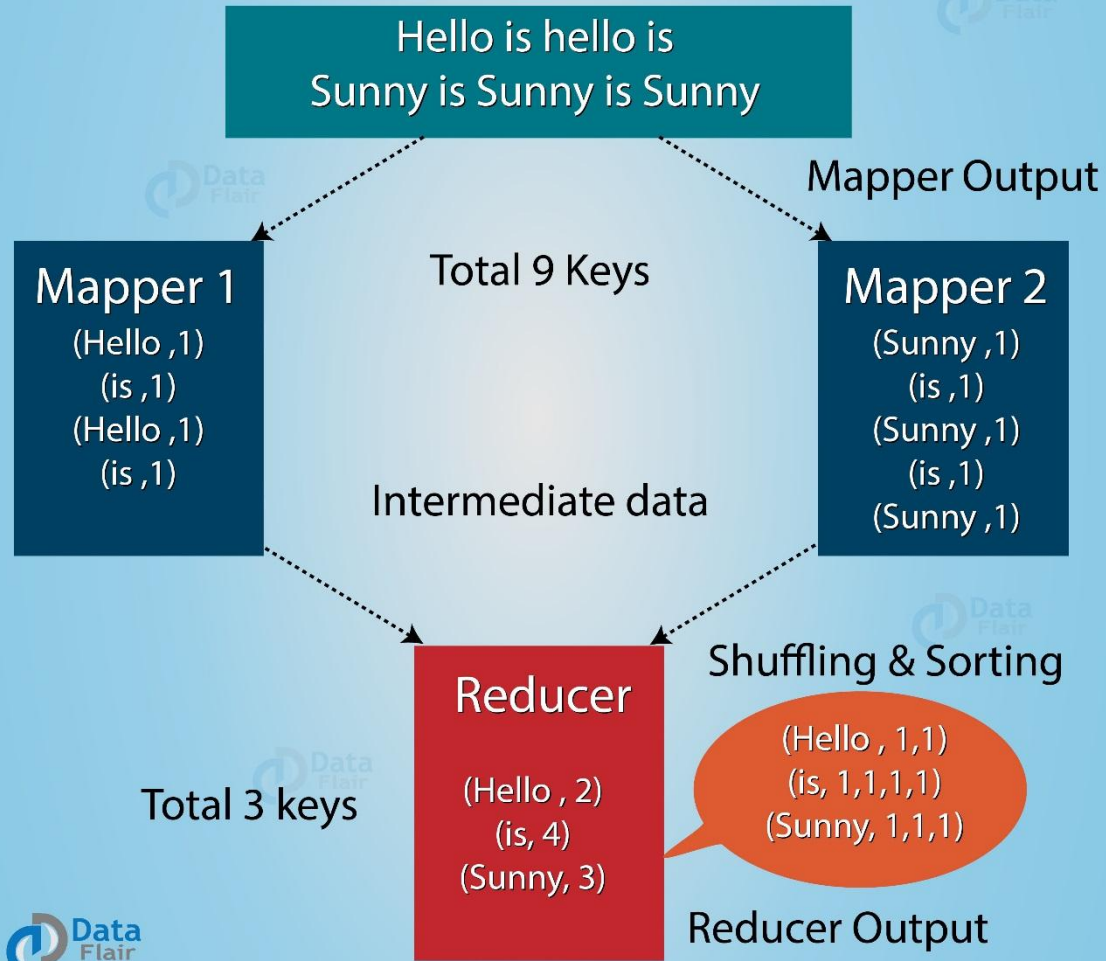


WHAT IS HADOOP COMBINER?

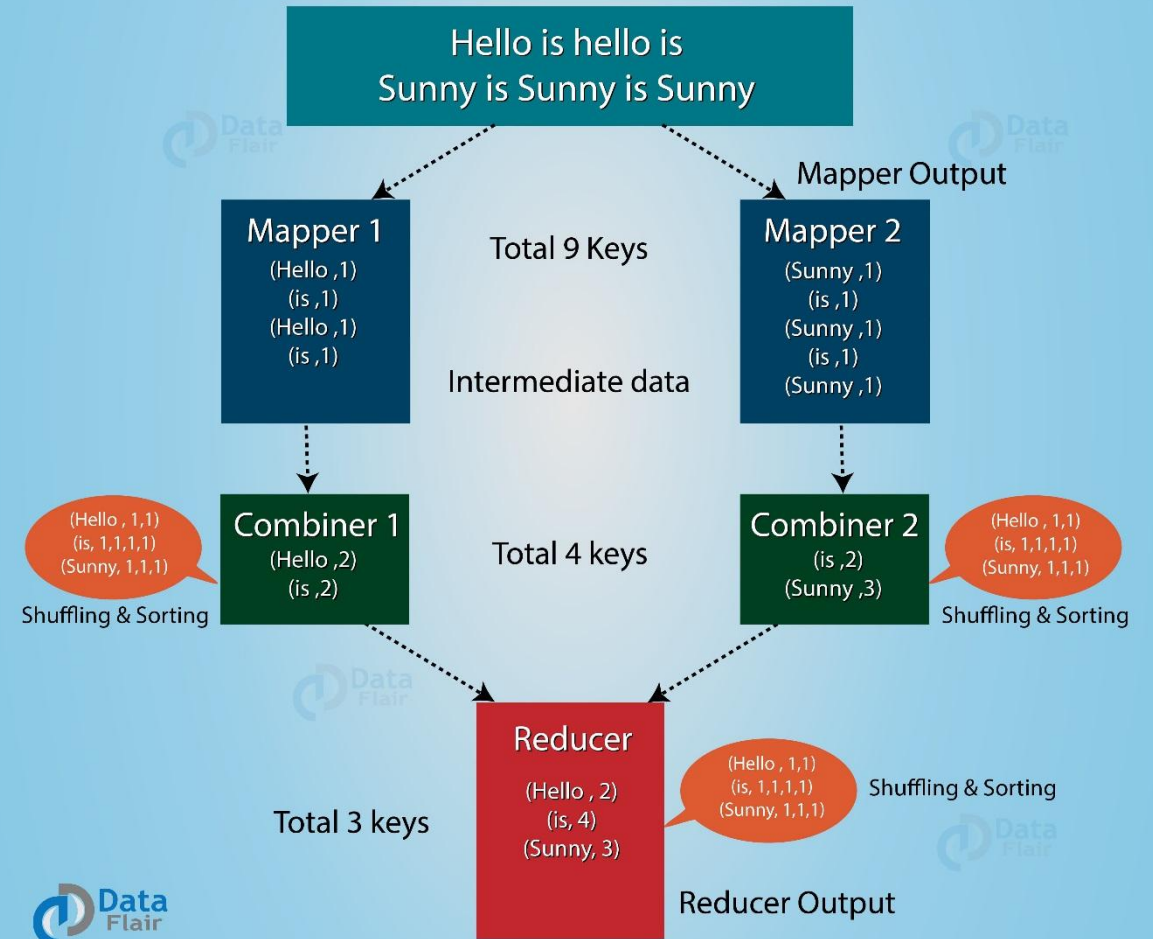
On a large dataset when we run MapReduce job, large chunks of intermediate data are generated by the Mapper and this intermediate data is passed on the Reducer for further processing, which leads to enormous network congestion. MapReduce framework provides a function known as Hadoop Combiner that plays a key role in reducing network congestion.

The combiner in MapReduce is also known as 'Mini-reducer'. The primary job of Combiner is to process the output data from the Mapper, before passing it to Reducer. It runs after the mapper and before the Reducer and its use is optional.

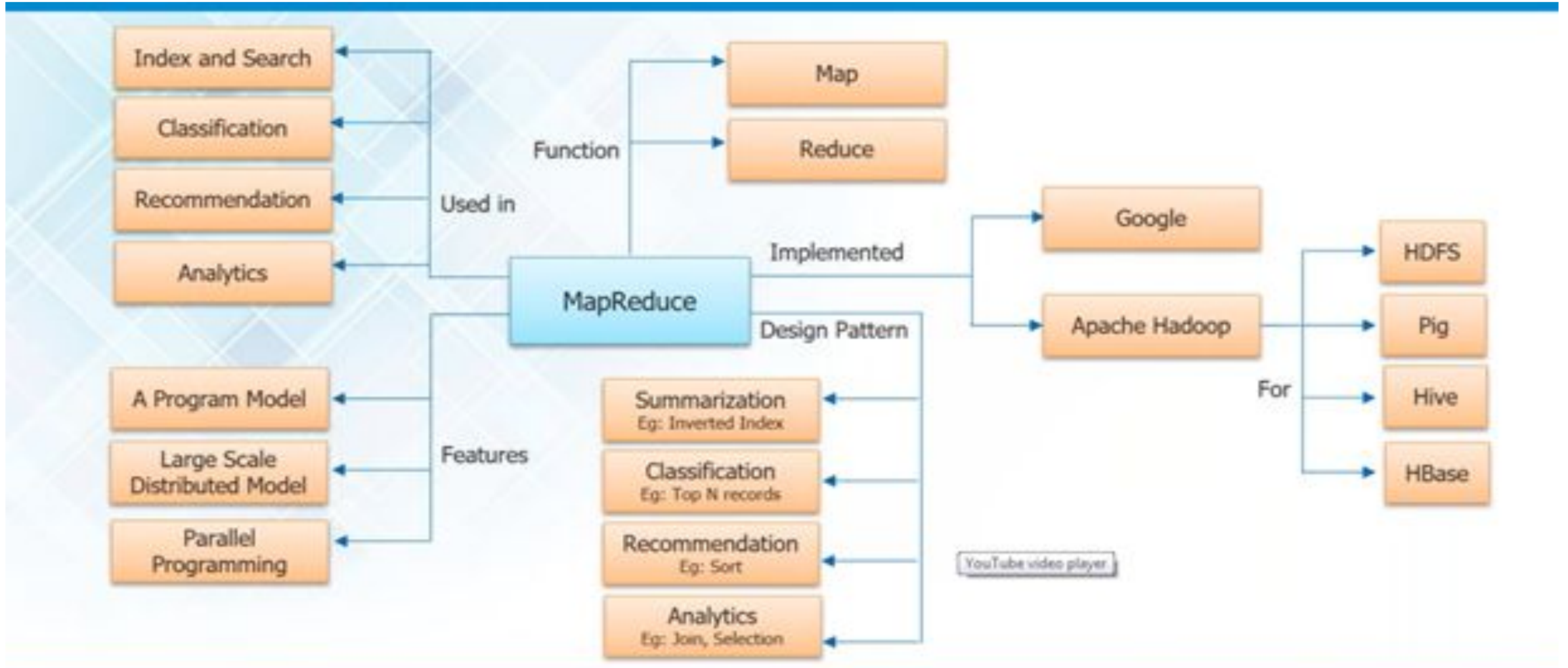
MapReduce program without Combiner



MapReduce program with Combiner



MAPREDUCE IN NUTSHELL



COPING WITH NODE FAILURE

1) Map TaskTracker failure:

- Map task completed or in-progress at TaskTracker, are reset to idle on failure
- Reduce TaskTracker gets a notice when a task is rescheduled on another TaskTracker.

2) Reduce TaskTracker failure:

- Only in-progress tasks are reset to idle

3) Master JobTracker failure:

- Map-Reduce task aborts and notifies the client.

IMPORTANT TOPICS

The function of the Map task within the MapReduce framework

Define the term "Reducer" in the context of MapReduce.

The concept of "Grouping by Key" in MapReduce.

Explain the concepts of union and intersection in the context of relational algebra operations.

What is Grouping & Aggregation in Relational Algebra Operations

Identify and describe a real-world example where MapReduce is effectively applied

Analyze and list the benefits associated with using the MapReduce process.



Analyze how relational algebra operations are computed and executed using MapReduce, including the role of Map and Reduce functions

Describe Map Reduce algorithm for Natural Join of two relations with example.

Describe Map Reduce algorithm to perform Intersection of two sets with example.

Evaluate matrix vector multiplication by map reduce. Illustrate with an example Consider the following matrix.

Explain the operation of grouping and aggregation using Map Reduce with Given example?

Demonstrate the Map Reduce program without combiner and with Combiner using suitable example

Create a detailed diagram that summarizes the MapReduce process in nutshell

Explain the operation of union using Map Reduce with Given example



the role of combiners in optimizing MapReduce jobs and provide an example.

Develop a pseudo code for implementing the Matrix-Vector Multiplication algorithm using MapReduce.

Compare and analyze traditional approaches to parallel and distributed processing, highlighting their methods and limitations.

the detailed execution process of a MapReduce job, from the submission to the output.

Analyze how MapReduce can be used to perform matrix-vector multiplication.

Illustrate with an example how MapReduce can be used to compute the projection operation in relational algebra.

Analyze strategies for coping with node failure in distributed computing environments.